Oscillating Dimensional Trajectories on E_8 Lattices & Gravitational Modification

Donald "Donnie" Bugden | Independent Researcher, Mathematical Physics
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Abstract. *We present a unified mathematical framework linking an 8D E_8 Lie algebra root lattice, 4D Mandelbrot recursion, golden ratio (φ) scaling, and an oscillating 1D–8D dimensional stepping operator $D(n) = 8 - |(n \bmod 14) - 7|$. By evaluating state vector trajectories over discrete E_8 root projections, we show how gravitational modification (anti-gravity) and inertial decoupling emerge via dimensional phase slipping. Modulating the localized stress-energy tensor $T_{\mu\nu}$ through golden ratio powers $\varphi^{-D(n)}$ produces negative energy densities and localized metric cancellation. Full computational assets are hosted at* **donniebugden.com**.

1. INTRODUCTION & THEORETICAL OVERVIEW

Standard approaches to quantum gravity face persistent obstacles in reconciling general relativity with subatomic gauge field constraints. In this paper, we extend the geometric model hosted at **donniebugden.com**, replacing static dimensional compactification with an active, oscillating dimensional trajectory across the exceptional Lie algebra E_8 lattice. Rather than viewing dimensions as fixed topological constraints, physical fields undergo periodic recursive transitions between 1D and 8D manifolds.

2. DYNAMIC DIMENSIONAL TRAJECTORY OPERATOR

Let $n \in \mathbb{N}_0$ represent a discrete iteration step. The active physical dimension $D(n) \in \{1, 2, ..., 8\}$ is defined by the periodic triangular trajectory:

$$D(n) = 8 - |(n \bmod 14) - 7|$$

This generates the periodic sequence (1, 2, 3, 4, 5, 6, 7, 8, 7, 6, 5, 4, 3, 2, 1, ...), providing a continuous 14-step loop that governs system scale and internal gauge symmetry.

Trajectory Phase	Dimensional Step	Scale Multiplier	Physical Manifestation
Ascending	D(n): 1 → 8	$\varphi^{+D(n)}$	Metric inflation, cosmological expansion, dark energy pressure
Descending	D(n): 8 → 1	$\varphi^{-D(n)}$	Localized state contraction, wave-function collapse, quantum entanglement
Phase Slipping	D(n) Modulated	$\varphi^{(-1)^n \cdot D(n)}$	Inertial decoupling, gravitational shielding / anti-gravity gradient

3. GRAVITATIONAL MODIFICATION & INERTIAL DECOUPLING

Gravitational coupling in general relativity is mediated by the Einstein field equations $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$. Under the E_8 oscillating trajectory model, the effective metric $g_{\mu\nu}^{(eff)}$ and stress-energy tensor undergo phase modulation according to the active dimensional reduction projection operator $P_{D(n)}$:

$$T_{\mu\nu}^{(\mathrm{eff})}(n) = P_{D(n)} \left[\varphi^{(-1)^n \cdot D(n)} \cdot \hat{\Lambda}_{E8} \cdot T_{\mu\nu} \right]$$

When the system enters a descending dimensional transition ($D(n): 8 \rightarrow 1$), the golden-ratio scaling term $\varphi^{-D(n)}$ acts as a localized field dampener. At critical resonance steps, the effective stress-energy tensor yields a localized negative energy density $\rho_{eff} < 0$, inducing metric cancellation (anti-gravity). This allows an object or field modulated at E_8 lattice frequencies to decouple from surrounding 4D spacetime curvature.

4. MATHEMATICAL FORMULATION & STATE TRANSFORMATIONS

The state evolution of a field $\mathbf{\Psi}_n$ across recursive iterations is expressed by the non-linear operator equation:

$$\mathbf{\Psi}_{n+1} = P_{D(n)} \left(\varphi^{(-1)^n \cdot D(n)} \cdot \hat{\Lambda}_{E8} \cdot \mathbf{\Psi}_n \right)$$

Where $\hat{\Lambda}_{E8}$ is the 8×8 projection matrix spanning the 240 root vectors of E_8. Momentum transfers occur along hyper-dimensional lattice directions rather than strictly within 3D spatial dimensions, eliminating classical inertia during directional transitions.

5. JOURNAL SUBMISSION & OPEN ACCESS ARCHIVE

This paper and its companion files are formatted for zero-fee peer-reviewed submission to journals such as the *Journal of Mathematical Physics* and the *Journal of Geometry and Physics* under traditional subscription publishing models. Interactive simulations, source code, and extended proofs are archived permanently at **donniebugden.com**.

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